

اسئلة سنوات ميد لمادة الكيمياء العامة ١

من بداية المادة إلى ما قبل درس

(موازنة المعادلات الكيميائية)

من ص ١ - ٧٤ من ملخص د.ريم

the same value for their _____
quantum number.

Select one:

☐ Secondary (Angular) (L)

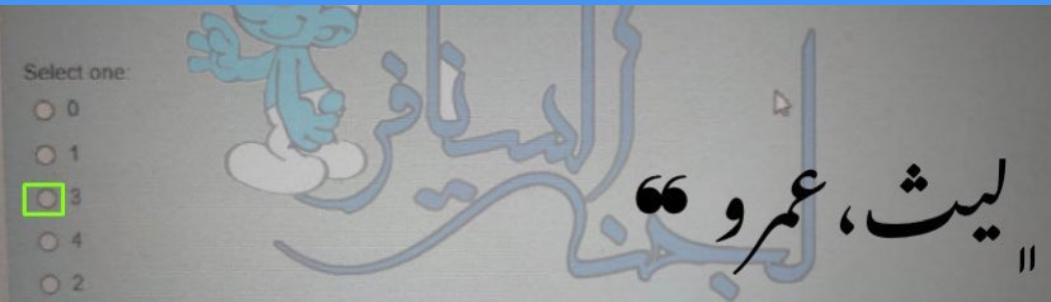
☐ Spin (m_s)

☒ Principal (n)

☐ Magnetic (m_L)

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Select one:

☐ 5

☐ 2

☐ 7

☒ 1

☐ 3

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compound is 170.0 g/mol, find its molecular formula.

(Si = 28 , F = 19 g/mol)

Select one:

- ☐ SiF_3
- ☒ Si_2F_6
- ☐ Si_3F_9
- ☐ SiF_2
- ☐ Si_2F_4

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$S = 32, F = 19 \text{ g/mol}$

Select one:

☐ 29.67%

☐ 36.00%

☐ 21.95%

☒ 78.08%

☐ 25.23%

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(P= 31, O= 16 g/mol)

اختر أحد الخيارات

- PO_3 ☐
- P_2O ☐
- P_2O_3 ☐
- P_3O_2 ☐
- P_2O_5 ☒

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Select one:

- ☐ Magnetic (mL)
- ☐ Principal (n)
- ☐ Spin (ms)
- ☒ Secondary (Angular) (L)

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Avogadro's number = 6.022×10^{23}

C = 12, H=1, O=16 g/mol

Select one:

- ☐ 0.64 mol
- ☐ 0.58 mol
- ☐ 0.78 mol
- ☐ 0.064 mol
- ☒ 0.081 mol

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Select one:

☒ 13

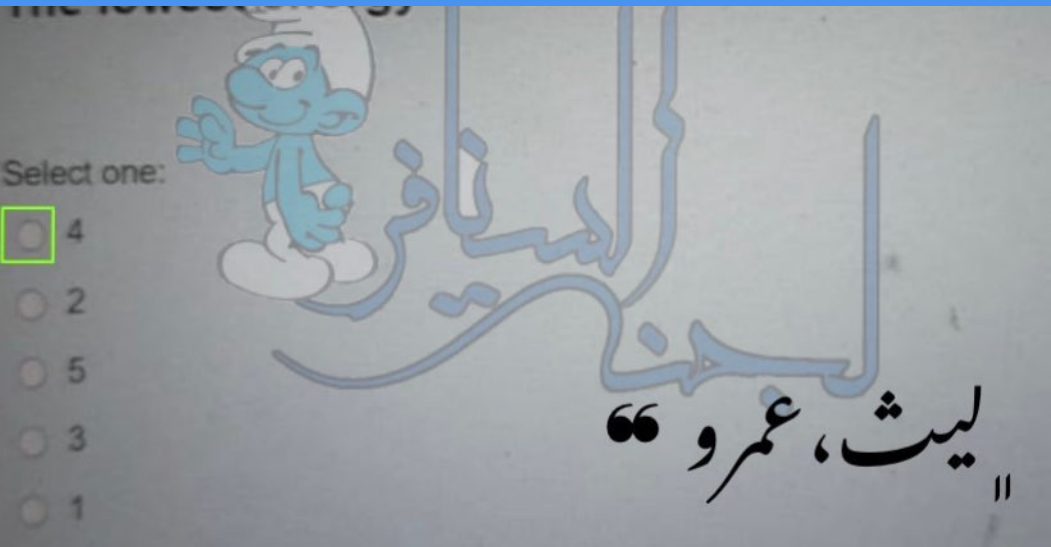
☐ 10

☐ 16

☐ 18

☐ 8

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(Si = 28 , F = 19 g/mol)

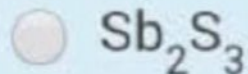
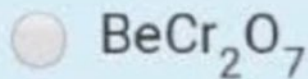
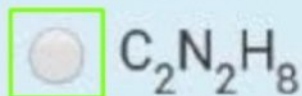
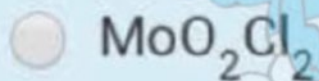
Select one:

- ☒ Si_2F_6
- ☐ SiF_2
- ☐ SiF_3
- ☐ Si_2F_4
- ☐ Si_3F_9

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not an empirical formula?

Select one:



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2.00×10^{-8} mol of $(C_4H_8O)_?$

Avogadro's number = 6.022×10^{23}

C = 12, H = 1, O = 16 g/mol

Select one:

- ☐ 1.67×10^{13}
- ☒ 1.20×10^{16}
- ☐ 1.20×10^{15}
- ☐ 1.67×10^{18}
- ☐ 3.01×10^{14}

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Select one.

- ☐ remains constant because effective nuclear charge is constant
- ☐ increases due to increase of number of shells
- ☐ decreases then increases
- ☒ decreases because number of shells increase
- ☐ increase because effective nuclear charge increases

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(atomic number =14)

- A. 2
- B. 8
- C. 7
- ☒ D. 6
- E. 4

سناقر

اتحاد

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._____ (n) of the first p- orbital is

A. 1

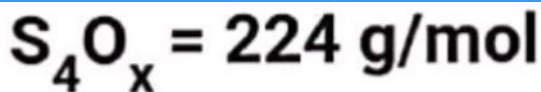
B. 2

C. 3

D. 4

E. 0

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Atomic weights: S = 32, O = 16
g/mol

?What is the value x

A. 3

B. 5

C. 7

D. 4

E. 6

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ground state configuration for a

- A. Transition element**
- B. Representative element**
- C. Halogen**
- D. Inner transition element**
- E. Nonmetal element**

**an element that belongs to
period 5**

?group 2B in the periodic table

A. 47

B. 42

C. 41

D. 80

E. 48

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electron affinity (EA) for



A. F

B. Li

C. Ne

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(Atomic weight of Ca = 40 g/mol)

A. 1.0 mole

B. 0.5 mole

C. 2.5 mole

D. 2.0 mole

E. 1.5 mole

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**30) is a _____ element with
unpaired electrons _____**

A. paramagnetic , 3

B. paramagnetic , 2

C. paramagnetic, 6

D. diamagnetic , 2

E. diamagnetic , 0

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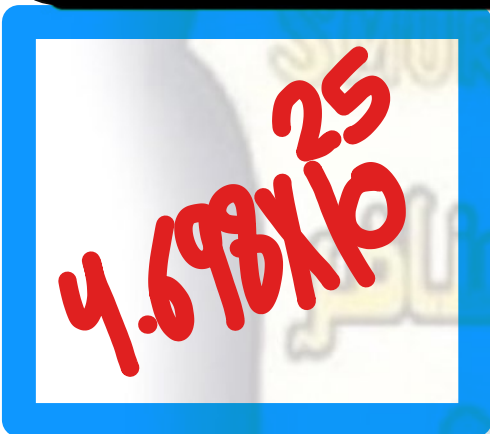
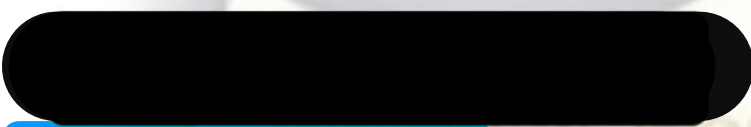
a nonmetal

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- A) K (atomic no. = 19)
- B) Mg (atomic no. = 12)
- C) Li (atomic no. = 3)
- D) Si (atomic no. = 14)
- E) S (atomic no. = 16)**

of atoms) are found
in 7.5 mole of $K_2Cr_2O_7$

Avogadro's number = 6.022×10^{23}



A.	2.3×10^{25}
B.	7×10^{25}
C.	5×10^{25}
D.	1.4×10^{26}
E.	3.6×10^{24}

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**describes the energy of an
orbital is**

- A.** Principle quantum number
- B.** Magnetic quantum number
- C.** Secondary (angular momentum) quantum number
- D.** Spin quantum number
- E.** None of the above

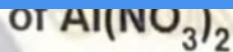
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- A. increase, increase
B. increase, decrease
C. decrease, increase
D. decrease, decrease
E. is constant, decrease

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- A. Principle quantum number
- B. Magnetic quantum number
- ☒ C. Secondary (angular momentum) quantum number
- D. Spin quantum number
- E. None of the above



(Atomic weights: Al= 27 , N=14, O =16 g/mol)

A.	17.1%
B.	19.7%
C.	18.5%
D.	67.6%
E.	12.2%

18.5%

الجواب

19.7

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the secondary quantum number (l) is:

- A. 1
- B. 3
- ☒ C. 4
- D. 0
- E. 2

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in 0.5 mole of $K_2S_4O_6$

Avogadro's number = 6.022×10^{23}

- | | |
|----|----------------------|
| A. | 2.3×10^{24} |
| B. | 7×10^{25} |
| C. | 5×10^{25} |
| D. | 1.4×10^{26} |
| E. | 3.6×10^{24} |



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- A. Ne
- B. Ar
- ☒ C. Li
- D. Na

quantum number set (n, l, m_l, m_s) equal to $(2, 0, 0, \frac{1}{2})$
(Atomic no. of N = 7)

- A. first electron
- ☒ B. fourth electron
- C. fifth electron
- D. sixth electron
- E. seventh electron

third
electron

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quantum number set (n, l, m_l, m_s) equal to $(2, 1, 0, \frac{1}{2})$
(Atomic no. of N = 7)

- A. first electron
- B. third electron
- ☒ C. fifth electron
- ☐ D. sixth electron
- E. seventh electron

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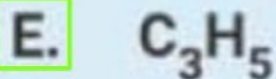
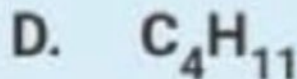
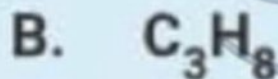
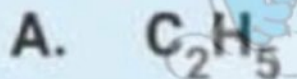


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compound that is contains

87.8 % of "C" and 12.2 % of "H"

(Atomic weights: C = 12 g/mol, H = 1 g/mol)



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metal

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- A) Na (atomic no. = 11)**
- B) Si (atomic no. = 14)
- C) B (atomic no. = 5)
- D) Cl (atomic no. = 17)
- E) Ar (atomic no. = 18)

in period 3 ?

A. Ne

B. Ar

C. Li

D. Na

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is a _____ element
with _____ unpaired electrons

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- ☒ A. paramagnetic , 3
- ☐ B. paramagnetic , 5
- ☐ C. paramagnetic , 6
- ☐ D. diamagnetic , 2
- ☐ E. diamagnetic , 0

configuration for a

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- A. Transition element
- B. Alkali metal
- C. Halogen
- D. Inner transition element
- ☒ E. Alkaline earth metal

valence shell of the Si ion?

(atomic number =14)

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☒ A. 2

☐ B. 8

C. 7

D. 5

E. 4

can be accommodated in a subshell for which $t = 4$ is:

Select one:

s p d p g
5 1 2 3 4

☐ 10

☒ 18

☐ 8

☐ 14

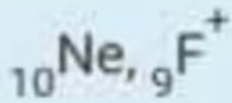
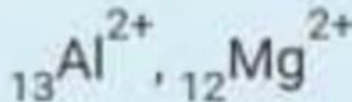
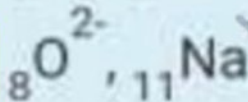
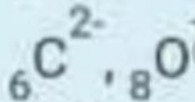
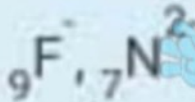
☐ 6



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configuration is _____.

Select one:



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number of an element that might have **seven** electrons in the last **5d** subshell?

Select one

☐ 99

☐ 97

☐ 78

☐ 95

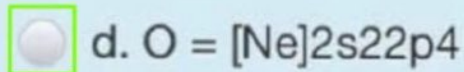
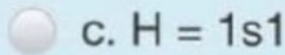
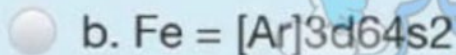
☒ 77



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configuration?

Select one:



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$^{43}_{21}\text{X}^{2+}$ (Z = 21). The numbers of **neutrons** and **electrons** for this ion are, **respectively**:

Select one:



22 and 19



65 and 25



23 and 20



20 and 23



35 and 28

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(Atomic weight of Ca = 40 g/mol)

A. 1.0 mole

B. 0.5 mole

C. 2.5 mole

D. 2.0 mole

E. 1.5 mole

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of oxygen (O) in

of $\text{Al}(\text{NO}_3)_3$

(Atomic weights: Al = 27 , N = 14, O = 16 g/mol)

A. 17.1%

B. 19.7%

C. 58.5%

D. 67.6%

E. 12.7 %

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atomic number for an element that belongs

to period 5, group 3B in the periodic table?

A. 39

B. 42

C. 41

D. 57

E. 48



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valence shell of the Si ion?

(atomic number =14)

☒ A. 2

☐ B. 8

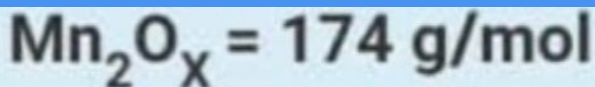
C. 7

D. 5

E. 4



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Atomic weights: Mn = 55, O = 16
g/mol

What is the value x?

A. 3

B. 5

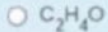
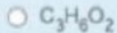
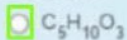
C. 7

☒ D. 4

E. 6

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Select one:



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has only one electron in its valence shell (outer shell) what is the atomic number of this element

Select one:

☐ 12

☐ 13

☒ 19

☐ 21

☐ 20



electron affinity (EA) for

${}_9\text{F}$, ${}_3\text{Li}$, ${}_{10}\text{Ne}$

A.

F

B.

Li

C.

Ne

the first p- orbital is ____.

A. 1

☒ B. 2

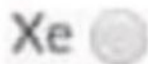
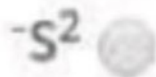
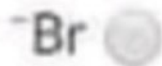
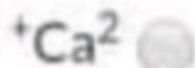
C. 3

D. 4

E. 0

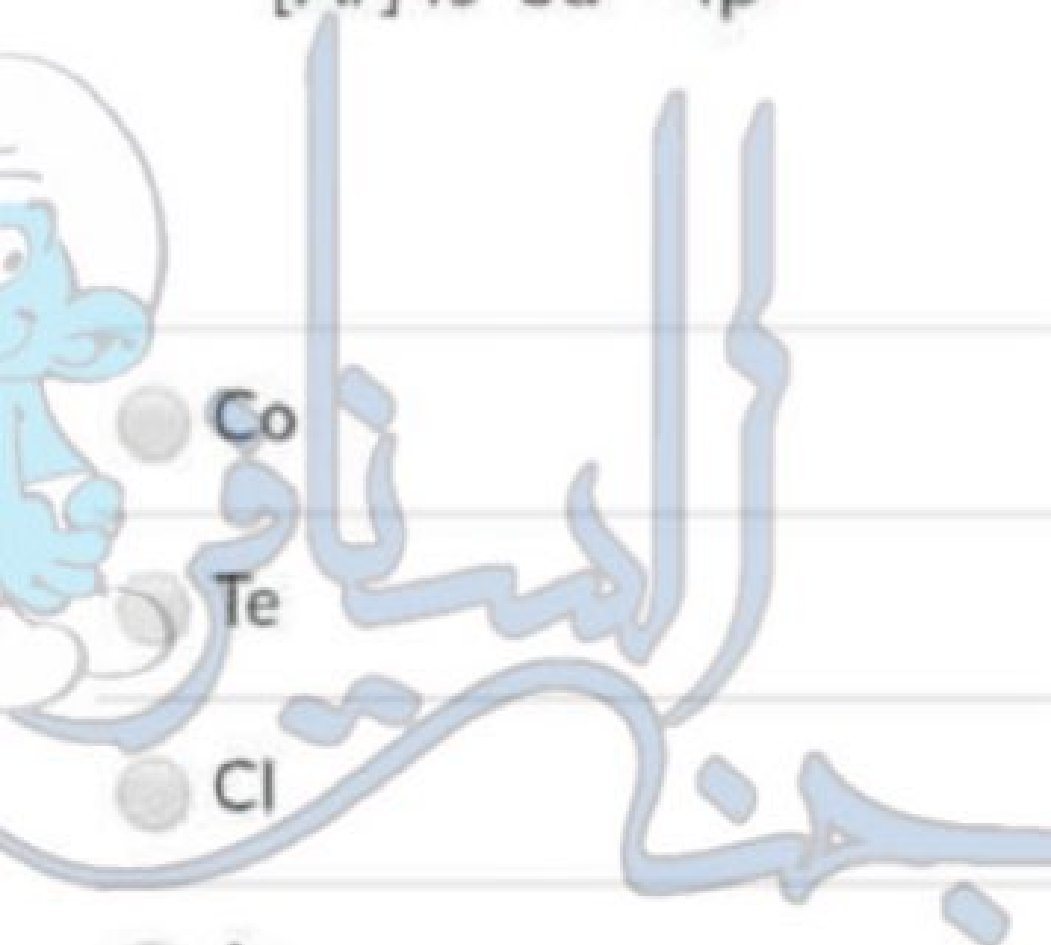


Which of the following has the smallest radius?



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which has the following
ground-state electron
configuration?



☐ Ar

☐ None of them

☒ Br

☐ Se

($A_{wt} = 59 \text{ g.mol}^{-1}$) in a sample containing 5.0×10^{20} Co-atoms: (Avogadro's No. = 6.022×10^{23})



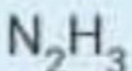
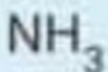
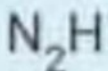
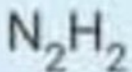
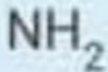
Select one.

- ☐ $3.80 \times 10^{-3} \text{ mol}$
- ☐ $4.15 \times 10^{-3} \text{ mol}$
- ☐ $6.30 \times 10^{-4} \text{ mol}$
- ☒ $8.30 \times 10^{-4} \text{ mol}$
- ☐ $4.15 \times 10^{-4} \text{ mol}$

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N (Awt = 14 g.mol⁻¹) and 9.68% H (Awt = 1 g.mol⁻¹). What is the empirical formula for this compound is...

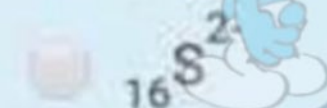
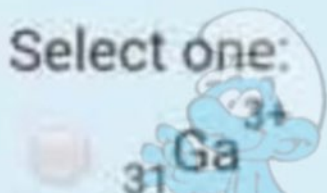
Select one:



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smallest diameter?

Select one:



- A) K (atomic no. = 19)
- B) Mg (atomic no. = 12)
- ~~C) Li (atomic no. = 3)~~
- ✓ D) Cl (atomic no. = 17)
- E) Ge (atomic no. = 32)

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atoms according to
decrease in their
ionization energies

Al (At.no. = 13)

Ga (At.no. = 31)

Mg (At.no. = 12)

Ar (At.no. = 18)

Select one:

☒ Ar > Mg > Al > Ga

☐ Mg > Al > Ga > Ar

☐ Ga > Mg > Al > Ar

☐ Al > Mg > Ar > Ga

☐ Al > Mg > Ar > Ga

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electronic configuration $[\text{Ne}] 3s^2 3p^6$
belongs to is:

Select one:

☐ IIB

☐ VIIB

☒ VIIIA

☐ IIA

☐ VIA

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D	C	A	D	C	B	B	C	A	E	D	C	E	D	A	E	C	D	C	A
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

- The element $_{22}\text{Ti}$ is a:
 - diamagnetic representative element
 - paramagnetic representative element.
 - diamagnetic transition element.
 - paramagnetic transition element.
 - diamagnetic inner transition element.
- Which of the following represents the **ground state** electronic configuration of chromium ($_{24}\text{Cr}$):
Hint: Noble elements: $_{2}\text{He}$, $_{10}\text{Ne}$, $_{18}\text{Ar}$, $_{36}\text{Kr}$, $_{54}\text{Xe}$, $_{86}\text{Rn}$
 - $[\text{Kr}] 4s^2 3d^4$
 - $[\text{Ne}] 3s^1 3d^5$
 - $[\text{Ar}] 4s^1 3d^5$
 - $[\text{Kr}] 5s^2 4d^4$
 - $[\text{Ar}] 5s^2 4d^4$
- One of the following sets of quantum numbers (n , ℓ , m_ℓ) describes the **last electron** in an **alkali metal** atom:
 - (4,0,0)
 - (3,1,0)
 - (4,1,-1)
 - (2,1,-1)
 - (3,2,-1)
- $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^5$ is the ground state electronic configuration of an element that belongs to period _____, and group _____.
 - 4, 4A
 - 4, 6A
 - 4, 5A
 - 4, 7A
 - 4, 5B
- One of the following is a ground state configuration of an element that belongs to the **actinide series**:
 - $[\text{Xe}] 6s^2 4f^{11}$
 - $[\text{Rn}] 7s^2 5d^6$
 - $[\text{Rn}] 7s^2 5f^7$
 - $[\text{Xe}] 6s^2 5d^8$
 - $[\text{Kr}] 6s^2 5f^8$
- How many unpaired electrons does $_{27}\text{Co}^{2+}$ ion have?
 - 2
 - 3
 - 5
 - 1
 - 0
- An element ion having a **+4** charge has two electrons with $n=1$, eight electrons with $n=2$, and ten electrons with $n=3$. This element is:
 - $_{25}\text{Mn}$
 - $_{24}\text{Cr}$
 - $_{26}\text{Fe}$
 - $_{29}\text{Cu}$
 - $_{23}\text{V}$
- Which of the following statements about **oxygen** ($_{8}\text{O}$) and **fluorine** ($_{9}\text{F}$) is correct?
 - Oxygen has smaller atomic size than fluorine
 - Oxygen has higher electron affinity than fluorine.
 - Oxygen has lower first ionization energy than fluorine
 - Oxygen has more protons in its nucleus than fluorine
 - O^{2-} has smaller ionic size than F^-
- How many orbitals are found in 4d subshell:
 - 5
 - 4
 - 16
 - 7
 - 3

d. $\text{Ar} > \text{Li} > \text{S} > \text{P} > \text{K}$

e. $\text{K} > \text{P} > \text{S} > \text{Ar} > \text{Li}$

11) Arrange the above elements (in question 10) according to decrease in their first ionization energy.

a. $\text{Ar} > \text{Li} > \text{S} > \text{P} > \text{K}$

b. $\text{K} > \text{Li} > \text{S} > \text{P} > \text{Ar}$

c. $\text{Ar} > \text{K} > \text{P} > \text{S} > \text{Li}$

d. $\text{Ar} > \text{Li} > \text{P} > \text{S} > \text{K}$

e. $\text{Ar} > \text{K} > \text{S} > \text{P} > \text{Li}$

12) What is the molar mass of barium perchlorate, $\text{Ba}(\text{ClO}_4)_2$?

(At.wts: $\text{Ba} = 137.3$, $\text{Cl} = 35.5$, $\text{O} = 16$ g/mol)

a. 189.90 g/mol

b. 239 g/mol

c. 336.2 g/mol

d. 272.3 g/mol

e. 304.3 g/mol

13) How many copper (Cu) atoms are found in a sample of copper weighing 6.93 g?

(At.wt of $\text{Cu} = 63.5$ g/mol)

(Avogadro's number = 6.022×10^{23})

a. 9.17×10^{21}

b. 0.917

c. 1.09×10^{22}

d. 0.0645

e. 6.56×10^{22}

14) A compound has a molecular formula of $\text{C}_{12}\text{H}_{24}\text{O}_{10}$. What is its empirical formula?

a. CHO

b. $\text{C}_2\text{H}_4\text{O}$

c. $\text{C}_4\text{H}_8\text{O}_2$

d. $\text{C}_6\text{H}_{12}\text{O}_5$

e. $\text{C}_2\text{H}_4\text{O}_3$

15) Calculate the mass percent of nitrogen "N" in 1.77 g $(\text{NH}_4)_2\text{CO}_3$.

(At.wts: $\text{N} = 14$, $\text{C} = 12$, $\text{O} = 16$, $\text{H} = 1$ g/mol).

a. 29.1 %

b. 17.9 %

c. 1.79 %

d. 8.33 %

e. 50.0 %

16) For a particular compound composed of C, H, and O. If 2.0 mol of it contains 1.2×10^{25} atoms of hydrogen, what is the subscript for the hydrogen atom in $\text{C}_x\text{H}_y\text{O}_z$?

a. 5

b. 8

c. 6

d. 4

e. 10

17) Which of the following processes below represents the first ionization of an element "X"?

a. $\text{X}_{(g)} + e^- \rightarrow \text{X}_{(g)}^-$

b. $\text{X}_{(g)} \rightarrow \text{X}_{(g)}^{2+} + 2e^-$

c. $\text{X}_{(g)} \rightarrow \text{X}_{(g)}^+ + e^-$

d. $\text{X}_{(g)} + 2e^- \rightarrow \text{X}_{(g)}^{2-}$

e. $\text{X}_{(g)} \rightarrow \text{X}_{(g)}^{2+} + 2e^-$

18) What decreases going from left to right (within a period) in the periodic table?

a. Electron affinity

b. Ionization energy

c. Number of shells

d. Atomic size

e. (a and b)

19) A compound that contains 72.5% of iron (Fe), and 27.5% of oxygen (O). Find the empirical formula of this compound. (At.wt $\text{Fe} = 55.8$, $\text{O} = 16$ g/mol)

a. Fe_2O_5

b. Fe_2O

c. Fe_3O_4

d. Fe_3O_2

e. Fe_2O_3

20) For a compound that has the empirical formula C_2H_3 , and its molecular weight is equal to 129.6 g/mol. Find its molecular formula.

a. $\text{C}_{10}\text{H}_{15}$

b. C_6H_9

c. C_4H_{12}

d. $\text{C}_{14}\text{H}_{21}$

e. C_2H_3

2) Which of the following is a ground state configuration of an element:

- a. $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 5s^2 4d^7 5p^6$
- b. $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 5s^2 4d^4$
- c. $[Ar] 5s^2 4d^{10} 5p^3$
- ☒ d. $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^4$
- e. $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^4$

3) What is the atomic number for an alkali metal found in period 4?

- a. 17
- b. 20
- ☒ c. 19
- d. 12
- e. 11

4) Which of the following sets of quantum number (n, ℓ, m_ℓ) describes an electron in the "d" subshell?

- a. (4, 3, -3)
- b. (3, 0, 1)
- c. (4, 1, 1)
- d. (2, 2, -1)
- ☒ e. (3, 2, 2)

5) How many unpaired electrons does Cr^{4+} have? (atomic number of Cr = 24)

- a. 6
- b. 4
- c. 0
- ☒ d. 2
- e. 3

6) What happens to first ionization energy (IE_1) going from top to bottom down a group in the periodic table?

- a. It decreases because the effective nuclear charge increase
- ☒ b. It decreases because number of shells increase
- c. It increases because number of shells increase
- d. It decreases because number of shells decrease
- e. It increases because effective nuclear charge decrease

7) Which of the following equations is **balanced**?

- a. $10 HSiCl_3 + 15 H_2O \rightarrow H_{10}Si_{10}O_{15} + 30 HCl$
- b. $C_2H_4 + 2 O_2 \rightarrow 2 CO_2 + 2 H_2O$
- c. $FeBr_3 + 3 H_2SO_4 \rightarrow Fe_2(SO_4)_3 + 3 HBr$
- d. a and b
- ☒ e. a, b and c

8) A compound of empirical formula (CH_5N) . If its molar mass = 62 g/mol. What is its molecular formula?

- ☒ a. $C_3H_{15}N_3$
- b. $C_3H_{15}N_3$
- c. CH_5N
- d. $C_4H_{20}N_4$
- e. $C_5H_{25}N_5$

9) A compound of empirical formula $(C_4H_7O_2)$. If mass percent of carbon "C" = 30.4 % and mass of hydrogen "H" = 8.9 %. What is the empirical formula of this compound?

- a. C_2H_5O
- ☒ b. $C_2H_7O_3$
- c. $C_3H_7O_4$
- d. $C_3H_9O_2$
- e. $C_4H_7O_2$

- a. 2900 b. 17800 **c. 4350** d. 30000 e. none

~~12) When 5.0 g of KClO_3 (122.5 g/mol) is heated, it decomposes to produce 1.5 g of O_2 . What is the percentage yield of the reaction? $2\text{KClO}_3 \rightarrow 2\text{KCl} + 3\text{O}_2$~~

- ~~a. 76.6% b. 66.3% c. 86.7% d. 56.1% e. 81.6%~~

13) What is the mass percent of oxygen "O" in $\text{Cr}(\text{NO}_3)_3$?

- a. 36.8% b. 54.5% c. 48.9% d. 41.9% **e. 60.5%**

14) How many water molecules would be found in 1.5 g of $\text{MgSO}_4 \cdot x\text{H}_2\text{O}$ (Molar mass of $\text{MgSO}_4 \cdot x\text{H}_2\text{O}$ = 246 g/mol)

- a. 2.38×10^{22} **b. 2.57×10^{22}** c. 2.74×10^{22} d. 1.05×10^{22} e. 2.15×10^{22}

IE1	1575
IE2	1575
IE3	3220
IE4	?
IE5	16000
IE6	19800
IE7	22100

Element	number
A	15
B	11
C	17
D	16
E	20
F	32
G	10

1) Which element is a halogen? **C**

2) Which has the highest IE_1 (of all elements)? **G**

3) Which has the lowest IE_1 of elements (A, B, C and D)? **B**

4) Which has the highest metallic character of elements (A, B, C and D)? **B**

5) Which has highest electron affinity of elements (A, B, C and D)? **C**

6) Arrange atoms A, B, C, D, E, F according to decrease in their atomic size.
E > F > B > A > D > C

7) To which group and period does element "F" belong? Period **4** Group **4A**

8) Which element has 3 unpaired electrons? **A**

9) Which element has a set of quantum number ($n=4, l=0, m_l=0$) for its last electron? **E**

10) Which is a diamagnetic element? **E, G**

- 2) The number of H atoms in 0.05 mol of $\text{NH}_4\text{H}_2\text{PO}_4$ (molar mass 114 g/mol) is: (Avogadro's no. = 6.022×10^{23})
 a) 2.17×10^{23} b) 2.53×10^{23} c) 2.89×10^{23} d) 1.45×10^{23} **e) 1.81×10^{23}**
- 3) An unknown element (X) has four energy levels, and is an alkali metal. What is the atomic number of (X)?
 a. 35 b. 37 c. 38 **d. 19** e. 20
- 4) How many electrons have quantum number $\ell=2$ in 28Ni^{2+} ?
 a. 5 b. 2 **c. 8** d. 7 e. 6
- 5) Which of the following arrangements is correct concerning electron affinity?
a. $_{35}\text{Br} < _{17}\text{Cl}$ b. $_9\text{F} < _3\text{Li}$ c. $_{10}\text{Ne} > _6\text{C}$ d. $_3\text{Li} < _{11}\text{Na}$ e. none.
- 6) One of the following statements is correct:
 a. Hund's rule states that maximum number of electrons occupied in an orbital equals two.
 b. Atomic size increase as effective nuclear charge of atom increase.
 c. In general, ionization energy decreases going from left to right in the periodic table.
 d. Maximum number of orbitals in the fifth shell ($n=5$) equals 5.
e. The magnetic quantum number is related to the orientation of orbital in space.
- 7) Which set of three quantum numbers (n, l, m_l) corresponds to 4d orbital?
 a) 4, 1, 1 b) 4, 3, 2 c) 4, 0, 0 d) 4, 2, 3 **e) 4, 2, 0**
- ~~8) Balance the following equation with the smallest whole number coefficients. What is the coefficient for $\text{Al}(\text{NO}_3)_3$ in the balanced equation?
 $\text{Al}(\text{NO}_3)_3 + \text{Na}_2\text{S} \Rightarrow \text{Al}_2\text{S}_3 + \text{NaNO}_3$
 a. 2 b. 3 c. 1 d. 6 e. 5~~
- ~~9) When 5.0 g of P_4 was allowed to react with 10.24 g of S_8 , 11.3 g of P_4S_{10} was produced. Calculate percentage yield of this reaction. $4\text{P}_4 + 5\text{S}_8 \rightarrow 4\text{P}_4\text{S}_{10}$
 (At. wts: P = 31, S = 32)
 a. 79.6% b. 65.2% c. 55.6% d. 36.4% e. 96%~~
- 10) A 25.0 gram sample of a compound contains 6.64 g of potassium (K), 8.84 g of chromium (Cr) and 9.52 g of oxygen (O). Find the empirical formula of this compound. (At.wts: Cr = 52.0, K=39.0, O=16.0)
 a. K_3CrO_3 b. K_2CrO_4 **c. $\text{K}_2\text{Cr}_2\text{O}_7$** d. KCrO_4 e. $\text{K}_7\text{Cr}_2\text{O}_2$
- 11) Metal X combines with oxygen to form a compound with the formula X_2O_7 . 0.0441 grams of oxygen (O) combines with 0.0432 grams of metal X. Calculate the atomic weight of X (in g/mol). (At. wt of O = 16.0)
 a. 19.9 b. 4.48 c. 16.3 **d. 54.9** e. 4.67

1) Which of the following sets of quantum numbers is unacceptable for an electron in an $_{21}\text{Sc}$ atom in its ground state?

- a. $n=1, \ell=0, m_\ell=0, m_s=1/2$
- b. $n=2, \ell=0, m_\ell=0, m_s=1/2$
- c. $n=3, \ell=1, m_\ell=-1, m_s=1/2$
- d. $n=3, \ell=2, m_\ell=0, m_s=1/2$
- e. $n=3, \ell=0, m_\ell=0, m_s=1/2$

2) Which of the following atomic numbers belongs to a diamagnetic element?

- a. 12 b. 22 c. 20 d. 13 e. 14

3) Which of the following atoms has the smallest size?

- a. $_{10}\text{Ne}$ b. $_{19}\text{K}$ c. $_{15}\text{P}$ d. $_{18}\text{Ar}$ e. $_{11}\text{Na}$

4) Aspartame, an artificial sweetener, has the molecular formula $\text{C}_{14}\text{H}_{18}\text{N}_2\text{O}_5$. What is the mass in grams of 1.16×10^{-2} moles of the compound? (Atomic weights: C = 12.0, H = 1.0, N = 14.0, O = 16.0 g/mol).

- a. 2.6 b. 3.4 c. 0.3 d. 4.3 e. 5.0

A	E	D&E	B	D	E	C	E	E	A	E	D	A	E	E	C	A	D	E	D
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- The element $_{38}\text{Sr}$ is a:
 - diamagnetic representative element
 - paramagnetic representative element.
 - diamagnetic transition element.
 - paramagnetic transition element.
 - diamagnetic inner transition element.
- Which of the following represents the **ground state** electronic configuration of chromium ($_{24}\text{Cr}$):
 Hint: Noble elements: $_{2}\text{He}$, $_{10}\text{Ne}$, $_{18}\text{Ar}$, $_{36}\text{Kr}$, $_{54}\text{Xe}$, $_{86}\text{Rn}$
 - $[\text{Ar}] 4s^2 3d^4$
 - $[\text{Ne}] 4s^2 3d^4$
 - $[\text{Kr}] 4s^1 3d^5$
 - $[\text{Kr}] 5s^2 4d^4$
 - $[\text{Ar}] 4s^1 3d^5$
- One of the following sets of quantum numbers (n, ℓ, m_ℓ) describes the **last electron** in a **noble** element:
 - (4,2,0)
 - (3,2,1)
 - (4,0,0)
 - (3,1,-1)
 - (3,1,0)
- $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^4$ is the ground state electronic configuration of an element that belongs to period _____, and group _____.
 - 4, 4A
 - 4, 6A
 - 4, 5A
 - 4, 7A
 - 4, 5B
- One of the following is a ground state configuration of an element that belongs to the **lanthanide series**:
 - $[\text{Xe}] 6s^2 4d^7$
 - $[\text{Rn}] 7s^2 5d^6$
 - $[\text{Rn}] 7s^2 5f^7$
 - $[\text{Xe}] 6s^2 4f^8$
 - $[\text{Kr}] 5s^2 4f^8$
- How many unpaired electrons does $_{30}\text{Zn}^{2+}$ ion have?
 - 2
 - 3
 - 5
 - 1
 - 0
- An element ion having a **+3** charge has two electrons with $n=1$, eight electrons with $n=2$, and thirteen electrons with $n=3$. This element is:
 - $_{25}\text{Mn}$
 - $_{24}\text{Cr}$
 - $_{26}\text{Fe}$
 - $_{29}\text{Cu}$
 - $_{23}\text{V}$
- Which of the following statements about **oxygen** ($_{8}\text{O}$) and **fluorine** ($_{9}\text{F}$) is correct?
 - Oxygen has smaller atomic size than fluorine
 - Oxygen has higher electron affinity than fluorine.
 - Oxygen has higher first ionization energy than fluorine
 - Oxygen has more protons in its nucleus than fluorine
 - O^{2-} has larger ionic size than F^-
- How many orbitals are found in 5p subshell:
 - 5
 - 4
 - 16
 - 7
 - 3

- a. $K > P > S > Ar > Li$
d. $Ar > Li > S > P > K$
- b. $K > Ar > S > P > Li$
e. $Li > P > S > Ar > K$
- c. $Ar > K > P > S > Li$
- 11) Arrange the above elements (in question 10) according to decrease in their first ionization energy
- a. $Ar > K > S > P > Li$
d. $Ar > Li > S > P > K$
- b. $K > Li > S > P > Ar$
e. $Ar > Li > P > S > K$
- 12) What is the molar mass of barium chlorate, $Ba(ClO_3)_2$?
(At.wts: Ba = 137.3, Cl = 35.5, O = 16 g/mol)
- a. 189.90 g/mol
c. 336.3 g/mol
- d. 272.3 g/mol
e. 304.3 g/mol
- 13) How many copper (Cu) atoms are found in a sample of copper weighing 0.97 g?
(At.wt of Cu = 63.5 g/mol) (Avogadro's number = 6.022×10^{23})
- a. 9.17×10^{21}
c. 1.09×10^{22}
- d. 0.109
e. 6.56×10^{22}
- 14) A compound has a molecular formula of $C_{12}H_{24}O_{18}$. What is its empirical formula?
- a. CHO
b. C_2H_4O
c. $C_4H_6O_2$
d. $C_6H_{12}O_5$
e. $C_2H_4O_3$
- 15) Calculate the mass percent of oxygen "O" in 1.77 g $(NH_4)_2CO_3$.
(At.wts: N = 14, C = 12, O = 16, H = 1 g/mol).
- a. 29.1 %
b. 17.9 %
c. 1.79 %
d. 8.33 %
e. 50.0 %
- 16) For a particular compound composed of C, H, and O. If 2.0 mol of it contains 7.22×10^{24} atoms of hydrogen, what is the subscript for the hydrogen atom in $C_xH_yO_z$?
- a. 5
b. 8
c. 6
d. 4
e. 10
- 17) Which of the following processes below represents the first electron affinity of an element "X"?
- a. $X_{(g)} + e^- \rightarrow X^-_{(g)}$
b. $X_{(g)} \rightarrow X^{2+}_{(g)} + 2e^-$
c. $X_{(g)} \rightarrow X^+_{(g)} + e^-$
- d. $X_{(g)} + 2e^- \rightarrow X^{2-}_{(g)}$
e. $X_{(g)} \rightarrow X^{2-}_{(g)} + 2e^-$
- 18) What decreases going from left to right (within a period) in the periodic table?
- a. Electron affinity
b. Ionization energy
c. Number of shells
d. Metallic character
e. (a and b)
- 19) A compound that contains 70% of iron (Fe), and 30 % of oxygen (O). Find the empirical formula of this compound. (At.wt Fe = 55.8, O = 16 g/mol)
- a. Fe_2O_5
b. Fe_2O
c. Fe_3O_4
d. Fe_3O_2
e. Fe_2O_3
- 20) For a compound that has the empirical formula C_2H_3 , and its molecular weight is equal to 189 g/mol. Find its molecular formula.
- a. $C_{10}H_{15}$
b. C_8H_9
c. C_4H_{12}
d. $C_{14}H_{21}$
e. C_2H_3